# USE CASE SPECIFICATION for EXAM HALL SEATING ALLOCATION SYSTEM

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| Use Case ID: | EHSA4.0 | EHSA4.0 | EHSA4.0 |
| Use Case Name: | Generate Seating Allocation | Generate Seating Allocation | Generate Seating Allocation |
| Created By: | Abhishek Roy, Sahil Siddique | Last Updated By: | |
| Date Created: | 24-02-2026 | Date Last Updated: | |

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| Actors: | Admin (Primary), Student (Secondary), Invigilator (Secondary) |
| Description: | This use case enables the admin to automatically allocate students to exam hall seats while enforcing an 8-way adjacency constraint to prevent students with the same exam paper from sitting in any of the 8 surrounding positions (front, back, left, right, and 4 diagonals). The system generates seating allocations, publishes them, and enables students to search their seats and download admit cards. Invigilators can view seating charts for exam monitoring. |
| Preconditions: | Admin must be logged in. Hall data must exist with layout information (UC1.0 completed). Student list must be uploaded with paper codes (UC2.0 completed). Exam schedule must be created. System has necessary permissions to generate PDFs and send notifications. |
| Post conditions: | All students are allocated to seats (or flagged as unallocated if capacity insufficient). Seat assignments are saved in the database with status "allocated". Allocation is published and visible to students. Students can search seats (UC5.0) and download admit cards (UC6.0). Invigilators can view seating charts. Email/SMS notifications sent to all allocated students. |
| Normal Flow: | 1. Admin logs into system<br>2. Admin navigates to "Generate Allocation" module<br>3. Admin selects exam from dropdown [A-1]<br>4. Admin selects allocation mode (same paper OR mixed papers) [A-2]<br>5. System validates that UC1.0 (Manage Hall Data) is completed [A-3]<br>6. System validates that UC2.0 (Upload Student List) is completed [A-4]<br>7. Admin clicks "Generate Allocation" button<br>8. System retrieves halls with layouts from database<br>9. System retrieves students with paper codes for selected exam<br>10. System validates hall capacity against student count [A-5]<br>11. System creates seat grid for each hall based on shape (rectangular, L-shaped, T-shaped)<br>12. System executes allocation algorithm: |

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| | a. Sort students by paper code<br>b. For each hall, iterate through students<br>c. Find next available empty seat<br>d. Check 8-way constraint (no same paper in 8 surrounding positions) [A-6]<br>e. If constraint satisfied, assign student to seat and mark seat occupied<br>f. If constraint violated, try next available seat<br>13. System saves all allocations to database<br>14. System displays allocation summary (total allocated, unallocated count, hall utilization)<br>15. Admin reviews allocation results [A-7]<br>16. Admin clicks "Publish Allocation" (UC4.0)<br>17. System changes allocation status to "published"<br>18. System enables UC5.0 (Search Seat) for students<br>19. System enables UC6.0 (Download Admit Card) for students<br>20. System sends notifications to all allocated students<br>21. Invigilators can now access seating charts via their interface |
| Alternative Flows: | A-1: If no exam is selected, display error "Please select an exam" and return to step 3<br>A-2: If allocation mode is not selected, use default mode "same paper" and continue<br>A-3: If UC1.0 not completed (no halls configured), display error "Please configure halls first" and terminate use case<br>A-4: If UC2.0 not completed (no students uploaded), display error "Please upload student list first" and terminate use case<br>A-5: If total hall capacity < student count, flag students exceeding capacity as unallocated and continue. Display warning to admin in results.<br>A-6: If 8-way constraint is violated for current seat, try next available seat in same hall. If no valid seat found in current hall, move to next hall. If no valid seat in any hall, mark student as unallocated.<br>A-7: If admin is not satisfied with allocation, admin can click "Re-generate" which returns to step 3 and clears current allocation data (not yet published).<br>A-8: If admin cancels before publishing, allocation is saved as draft with status "unpublished" and students cannot access it yet. |
| Exceptions: | If no halls are configured (UC1.0 incomplete), terminate use case and display "No halls available. Please configure halls first". If student list is empty for exam (UC2.0 incomplete), terminate and display "No students enrolled for this exam". If allocation algorithm runs for more than 5 minutes (timeout), save partial results and display warning "Allocation timeout. N students allocated, M students pending". If database connection fails during save, rollback all allocations and display "Database error. Please try again". |
| Includes: | UC2.0: Upload Student List (prerequisite - must be completed before allocation), UC4.0: Public Allocation (publishing step after generation), UC5.0: Search Seat (enabled after publishing), UC6.0: Download Admit Card (enabled after publishing, included in UC5.0) |
| Priority: | Critical (Must Have) |
| Frequency of Use: | Regular (every exam session, typically 2-4 times per semester) |

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| Business Rules: | 1. 8-way constraint is mandatory and cannot be overridden for same paper codes<br>2. Hall capacity is a hard limit and cannot be exceeded under any circumstances<br>3. Students with special requirements (wheelchair, scribe) must be assigned ground floor halls with accessible seating<br>4. Alternate seating (one seat gap) is enforced when same paper mode is selected<br>5. Mixed paper allocation maximizes hall utilization but still enforces 8-way constraint<br>6. Allocation must complete within 5 minutes for up to 1000 students, otherwise timeout occurs<br>7. Seat assignments are immutable once published - only manual adjustment via separate use case is allowed<br>8. Students can only access UC5.0 (Search Seat) and UC6.0 (Download Admit Card) after allocation is published<br>9. Invigilators get read-only access to seating charts after publishing<br>10. Notifications (email/SMS) are sent automatically upon publishing, no admin intervention needed<br>11. Admit cards include QR code for verification at exam entrance |
| Assumptions: | All halls have rectangular base layout (L-shaped and T-shaped are marked extensions with blocked cells). Seat coordinates (row, column) are zero-indexed and pre-validated during hall configuration. Each student has exactly one paper code per exam, validated during upload. Paper codes are standardized across all exams and follow consistent naming convention. Special requirements are pre-validated and flagged in student master data. Database supports transaction rollback for failure scenarios. Email/SMS gateway is configured and operational. PDF generation library is available for admit card creation. Students have access to email and/or mobile phone for receiving notifications. |

**Iteration 1:**

Admin, Student, Invigilator, Hall, Exam, Seat, Allocation, Paper, Constraint, Validator, Controller, Algorithm, AdmitCard

// Allocation, Constraint, and Validator are control/process classes. Paper is an attribute of both Exam and Student. Controller and Algorithm are implementation classes. AdmitCard is a document generated from allocation data.

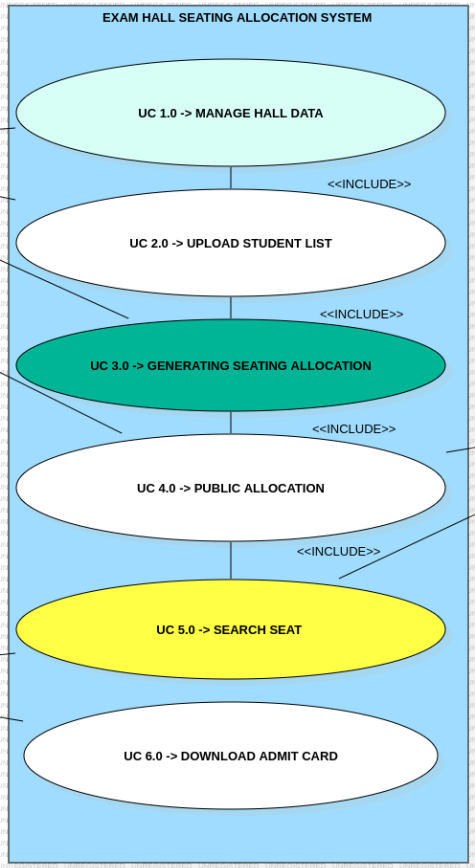
**Iteration 2:**

Admin, Student, Invigilator, Hall, Exam, Seat, SeatAllocation, AdmitCard

// SeatAllocation is the association class linking Student, Hall, Seat, and Exam. AdmitCard is generated document containing allocation details.

**Prospective Classes:**

Admin (adminId, name, email, role)
Student (studentId, rollNumber, name, course, paperCode, specialRequirements)
Invigilator (invigilatorId, name, assignedHalls[])
Hall (hallId, name, building, totalSeats, rows, columns, shapeType)
Exam (examId, examDate, examTime, courseCode, courseName)
Seat (seatId, hallId, seatNumber, row, column, isBlocked)
SeatAllocation (allocationId, examId, studentId, hallId, seatId, paperCode, status)
AdmitCard (admitCardId, studentId, examId, seatNumber, hallName, qrCode)


EXAM HALL SEATING ALLOCATION SYSTEM
UC 1.0 -> MANAGE HALL DATA
<<INCLUDE>>
UC 2.0 -> UPLOAD STUDENT LIST
<<INCLUDE>>
UC 3.0 -> GENERATING SEATING ALLOCATION
<<INCLUDE>>
UC 4.0 -> PUBLIC ALLOCATION
<<INCLUDE>>
UC 5.0 -> SEARCH SEAT
UC 6.0 -> DOWNLOAD ADMIT CARD
ADMIN
INVIGILATOR
STUDENT